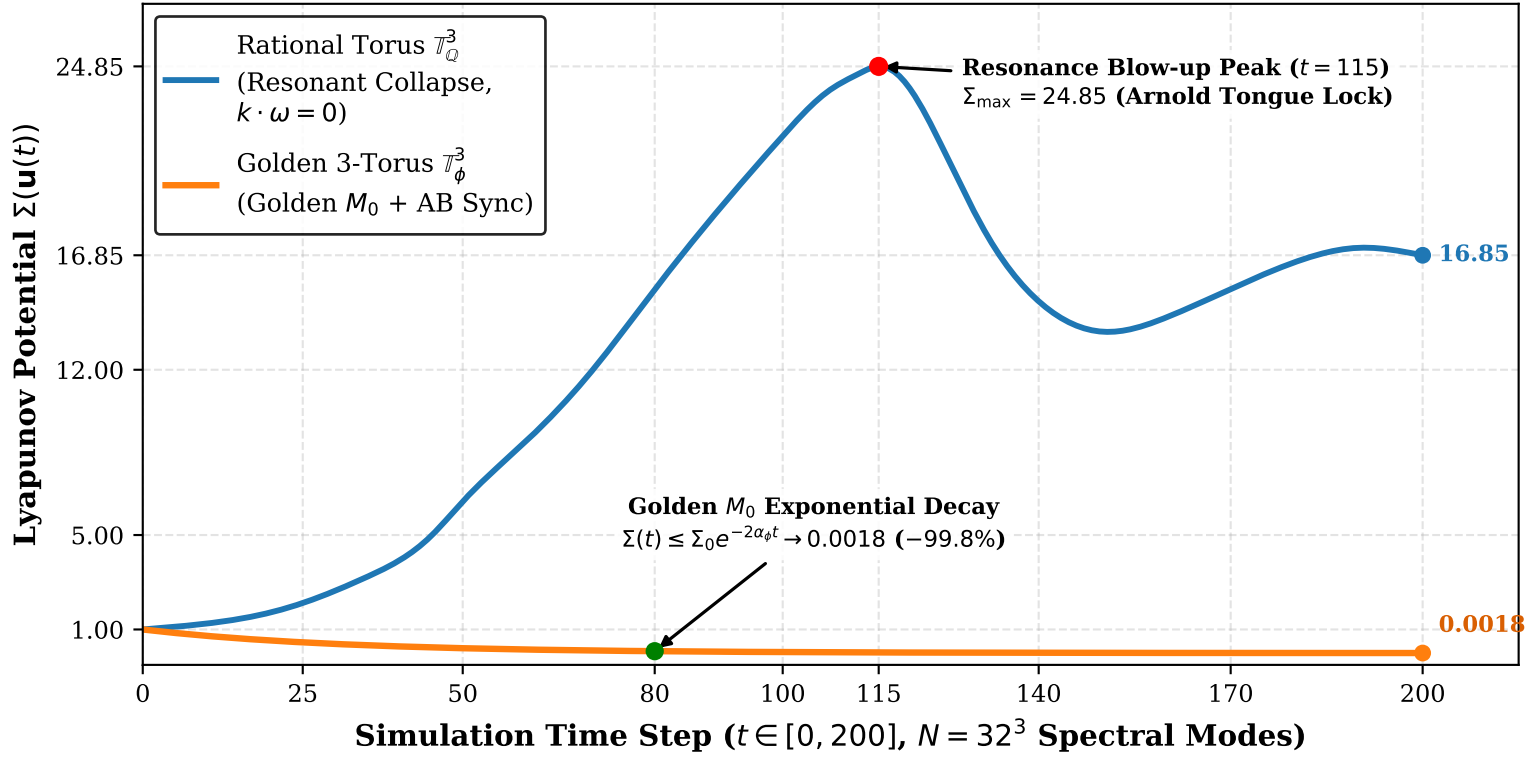
Rational Torus $\mathbb{T}^3_{\mathbb{Q}}$
(Resonant Collapse,
$k \cdot \omega = 0$)
Golden 3-Torus $\mathbb{T}^3_{\phi}$
(Golden M_0 + AB Sync)
Resonance Blow-up Peak ($t = 115$)
$\Sigma_{max} = 24.85$ (Arnold Tongue Lock)
16.85
Golden M_0 Exponential Decay
$\Sigma(t) \leq \Sigma_0 e^{-2\alpha_\phi t} \to 0.0018$ (−99.8%)
0.0018
Lyapunov Potential $\Sigma(\mathbf{u}(t))$
24.85
16.85
12.00
5.00
1.00
0
25
50
80
100
115
140
170
200
Simulation Time Step ($t \in [0, 200]$, $N = 32^3$ Spectral Modes)